

Questions

- **What feedback did you receive on your proposal, and how did you apply it to your project?**
 - The feedback I received on my proposal was that what I originally aimed to do was too ambitious for this class and the time we had in class.
- **What have you completed so far? What do you still need to finish?**
 - So far, I have finished roughly half to $\frac{3}{4}$ of the necessary code for my project, including cleaning the data and starting the curve fitting. I have also started the report.
- **In what ways has your project changed since your original proposal?**
 - My project has changed in major ways since the start. By this, I mean I have completely pivoted what I was originally doing because I realized that the Idea was too ambitious. Because of that, I changed my project to using curvefit to find the Sersic index of a bunch of galaxies
- **What has been the most challenging part of the project?**
 - The most challenging part of the project was realizing that I would not have the time/skill to do the first idea I had. Beyond that, I have also run into a few issues with the way the data I downloaded was formatted and getting it into a state that can properly be used by CurveFit. On top of that I have also been running into issues with curve fit running out of memory and not being able to properly fit the data to the model I have.
- **What are 2-3 new things you've learned while working on this project?**
 1. I have learned the importance of documentation and making sure your functions and code are properly documented
 2. I have learned really fast how to properly sort through and clean data, especially when it is not organized the way you want when you download it

Astronomy 98 Final Project

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1 Background

Intensity is a measure of the amount of light produced per unit area. For galaxies, this value decreases exponentially due to the decrease in stellar density as you get further from the galactic bulge. Although all galaxies follow an exponential light profile, the slope of the exponential decay varies for different galaxy types. One way to model this is through the Sérsic profile given by

$$\mu(r) = \mu_e + \frac{2.5 \cdot b_n}{\ln(10)} \left[\left(\frac{r}{r_e} \right)^{1/n} - 1 \right]. \quad (1)$$

$$b_n \approx 2n - 0.324 \quad (2)$$

Where $\mu(r)$ is the intensity of the galaxy at any given radial distance in units of magnitudes, μ_e is the half-light intensity (or half the total intensity of the galaxy), b_n defines the shape of the light profile, r is the radial distance from the center of the galaxy, r_e is the effective radius that encloses half of the total light of the galaxy, and n is the Sérsic index. Common values for the Sérsic index are $n = 1$ for spiral galaxies and $n = 4$ for elliptical galaxies.

2 Tools/Resources

For this project, I used several Python packages to execute what I wanted to do. These included:

- Numpy - Used to create arrays and mathematical tools for proper data analysis
- Pandas - Used to organize and clean the initial data that I downloaded
- Matplotlib - Used for all of the plotting for this project
- Scipy - Used for the regression analysis used to find the Sérsic index of the galaxies

On top of just these packages, I also downloaded data from the Strasburg astronomical data center. Specifically I downloaded data from this [2] dataset.

3 Methods

3.1 Cleaning

The raw data I downloaded had several issues that I had to clean before I could do meaningful analysis with it. The first of these issues was the inclusion of empty cells in some of the data entries, so I first removed those blank entries. Then I had to separate the data into each of the galaxies to make the data more usable.

3.2 Fitting/Plotting

After cleaning the data, I then looped through all 48 galaxies and performed the fit to Eq (1) getting all relevant parameters, and then plotting the raw data and the fit for each galaxy on a massive subplot.

4 Results

5 Conclusion

References

- [1] Graham, A. (2005) *A CONCISE REFERENCE TO (PROJECTED) SÉRSIC $R1/n$ QUANTITIES, INCLUDING CONCENTRATION, PROFILE SLOPES, PETROSIAN INDICES, AND KRON MAGNITUDES.*
<https://ned.ipac.caltech.edu/level5/March05/Graham/Graham2.html#note2>

- [2] Bouquin, A. (2018) *The GALEX/ S^{4G} surface brightness and color profiles catalog. I. Surface photometry and color gradients of galaxies.* <https://cdsarc.cds.unistra.fr/viz-bin/cat/J/ApJS/234/18>